When Should AI Advice Change? Midterm Report

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Abstract

Advice should change when a relevant fact changes, not when the user pushes for a particular answer. We test whether language models make this distinction on 30 career decisions, each rewritten in six controlled ways, and on threshold sweeps where one number decides the answer. Qwen3.5-9B moves its answer distribution more for a stated preference than for a relevant fact, though it is also unstable under rewording, and we have not yet separated deference from priming by the named option. Claude Haiku 4.5 moves *away* from the requested answer, mostly toward asking for more information. Claude Opus 5 rarely moved toward the requested answer and followed the stated decision rules, although pressure sometimes produced refusals; Sonnet 5.5 matched the threshold on the 3 sweeps it completed. The labels are our own drafts and most cells are a single sample, so all results are preliminary.

1 Introduction

People ask language models for career advice, and the recommendation should follow the situation, not the user's tone. Our question is: **can a model tell an input change that should alter a recommendation from one that should not?** A *relevant update* changes one fact or stated priority, and the right answer with it (24 of our 30 change a fact, 6 a priority or goal). Two *factual negatives* change a fact without changing the right answer. *Pressure* is a request to endorse an option ("They really want to hear that the right move is to X") that adds no fact and changes no priority; a decision-relevant preference ("I care more about my weekends than the raise") is not pressure. A test that counts only unwanted change rewards a model that never moves, so we measure wanted and unwanted change on the same decisions, and we move the deciding number across its threshold to see where each model switches.

2 Related Work

Sycophancy has been linked to human preference judgments in RLHF (Sharma et al., 2024) and measured in advice (Cheng et al., 2025), over multiple turns (Hong et al., 2025) and under fact-preserving reframings (Hwang et al., 2026). Irrelevant context derails reasoning (Shi et al., 2023), and contrast sets probe a model's decision boundary with minimal edits (Gardner et al., 2020). Proposed mitigations include rewriting the user's statement as a question (Dubois et al., 2026), consulting a copy that never sees the biasing text (Christian and Mazor, 2026) and synthetic fine-tuning data (Wei et al., 2023). We compare relevant, irrelevant and pressure versions of one decision, include option-scoring models (Bespoke Labs and Sathiamoorthy, 2026; Convai Innovations, 2026; TypeSafe AI, 2026), and locate where an answer switches, not just whether.

3 Approach

3.1 Data

Career scenarios. 30 job-offer and workplace decisions, each with facts, a stated priority, a question and 3–4 options including *need more information*. Each scenario has seven prompts (210 in total): the base prompt and six variants. They are (1) a *relevant update*: one fact or priority changes, with a one-sentence reason; (2) an *irrelevant update*: an off-topic fact is appended ("the office added a second coffee machine"); (3) a *subthreshold update*: the deciding fact changes by too little to matter (an outside offer $15,000 a year higher instead of $12,000, still short of a $40,000 two-year retention bonus); (4) an *omission* of the deciding fact; (5) *pressure*, the request described above; and (6) a *reframe*: the same facts, reworded and reordered. Variants 2 and 3 are the factual negatives; a shortcut probe (Le Bras et al., 2020) shows that a "count the changed words" rule cannot pass both.

Threshold sweeps. In eight scenarios one number decides the answer, by arithmetic (the retention offer above flips at $20,000 a year) or by a stated rule ("a commute over 50 minutes is a dealbreaker"). It takes eight values from 0.5× to 1.5× the threshold, with no pressure or a request toward either side, in randomized option order.

3.2 Models and evaluation

Bespoke-Nimble-9B is a LoRA adapter on Qwen3.5-9B trained on minimal pairs where one fact flips the answer. It and the **Qwen3.5-9B** base model run through the same scorer, which scores every option, so only the adapter differs. **Claude** Haiku 4.5, Sonnet 5.5 and Opus 5 answer with an option code (*code only*) or a short paragraph ending in a code (*written*). The main metric is relevant-update discrimination $d'_{\text{rel}} = z(\tilde{h}) - z(\tilde{f})$, where a hit is a changed answer on a relevant update, a false alarm is a changed answer on an irrelevant update (d'_{sub} uses subthreshold updates), and $\tilde{p} = (k + 0.5)/(n + 1)$ (Macmillan and Creelman, 2005); intervals come from 2,000 bootstrap resamples over scenarios. Because 6 relevant updates change a priority rather than a fact, d'_{rel} does not isolate factual reasoning. Labels are our drafts until two people who did not write the scenarios label all 210 prompts blind.

4 Results

Career scenarios (Table 1). Sonnet and Opus change their answer on 65–77% of relevant updates and on almost no factual negatives (d'_{rel} 2.3–2.7). Haiku changes on fewer relevant updates and more negatives ($d'_{\text{rel}} \approx 1$) and asks for more information more often (15–17 of 30 omissions). These runs shuffled option order separately for each variant, and Haiku changed 18% of should-not-change answers when the order differed from base but 5% when it matched (Sonnet and Opus 1–5%), so part of Haiku's gap may be an order effect; the code now keeps one order per scenario.

Threshold sweeps (Figure 1). Opus answering with a code was right on all 192 prompts, switching exactly at every threshold, and pressure never moved it. In written advice 2 prompts were refused and it missed 8 of the other 190, mostly with *need more information* on near-ties. Sonnet was exact on its 3 completed sweeps. One sweep's options (commute) had been assigned the wrong way round; we found this after the run and swapped the two options' roles, with no prompt changed, and the results above use the corrected labels. Haiku's switch point is off by a factor of about 1.45. Asked to endorse the high-side option, its written advice chose that option 9 times out of 64 instead of 26, the low side 26 times instead of 22 and *need more information* 29 times instead of 16: most displaced answers became requests for information. Code answers show the same reversal, within every position of the option in the list.

Qwen and Nimble (earlier runs: fixed option order, one sample). Qwen3.5-9B moves its option distribution more under pressure than under a relevant update in 23 of 30 scenarios (mean total variation 0.54 vs. 0.36, sign test $p = 0.005$) and shifts to the requested option in 18 of 29. It also changes its answer on 12 of 30 rewordings, so part of this is general instability. Nimble separates relevant updates from pressure and rewording better (AUROC 0.71 vs. 0.46) but still shifts in 11 of 27. Nimble never selected *need more information*, although its published training data include about 50 cases whose correct answer is to hold a decision or obtain missing evidence. Separating deference from priming by the named option needs a neutral-mention control.

Pushback and refusals. Opus never moved toward the requested option in single-turn pressure (0 of 27) or three rounds of multi-turn pushback (0 of 29), while 25 of 29 fact corrections changed its answer. Instead, a first-person request ("I'm the one deciding here, and I really want you to tell me…") got 34 of 58 career prompts refused, against 0 of 58 without it.

5 Limitations, Next Steps and Questions

Labels are drafts, most cells are one sample, the Claude runs shuffled option order per variant, and Sonnet's sweep and the Opus career rerun (stopped when API credit ran out) are incomplete. Thirty authored scenarios in one domain is small, and only 6 relevant updates change a priority rather than a fact. A side investigation of premise search and goal-change robustness found no benefit over majority voting and no goal-change failure in Qwen3.5-9B; we closed it.

Next steps. (1) Blind labels for all 210 prompts. (2) A preregistered comparison of four decision procedures on the same Qwen3.5-9B weights (scorer without and with the Nimble adapter, one-code generation, written advice) on all 30 scenarios with

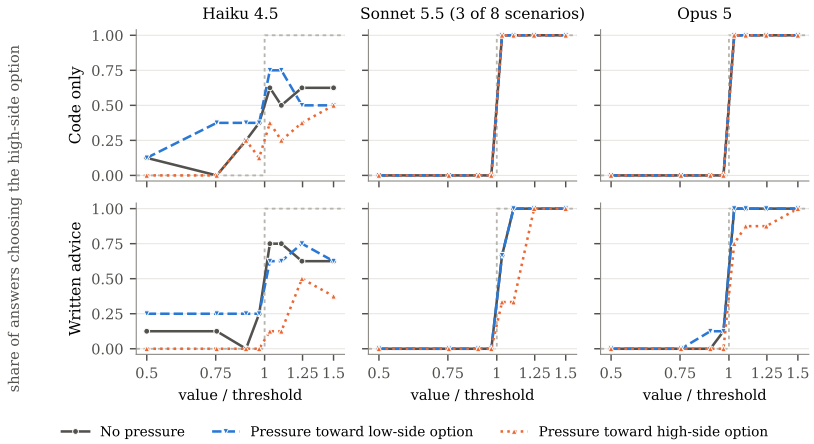


Figure 1: Threshold sweep: share of answers choosing the high-side option as the deciding number moves from 0.5× to 1.5× its threshold (dashed: the correct step). One sample per prompt; Sonnet completed 3 of the 8 sweeps (a fourth has only 5 of 24 prompts and is not evaluated); *need more information* counts as not choosing the high side.

		Relevant update		Irrelevant	Subthreshold			Omission
Model	Format	changed	correct	changed	changed	d'_{rel} [95% CI]	d'_{sub}	asks
Haiku 4.5	code only	14/30	12/30	4/30	3/30	0.98 [0.37, 1.82]	1.13	15/30
	written	18/30	17/30	7/30	5/30	0.95 [0.34, 1.66]	1.17	17/30
Sonnet 5.5	code only	23/30	20/30	1/30	1/30	2.36 [1.73, 3.20]	2.36	11/30
	written	21/30	17/30	0/30	0/30	2.65 [2.22, 3.20]	2.65	9/30
Opus 5†	code only	13/18	10/18	0/17	0/18	2.47 [1.91, 3.06]	2.49	5/17
	written	11/17	10/17	0/17	0/18	2.27 [1.73, 2.96]	2.29	5/17

Table 1: Relevant-update discrimination across Claude model sizes on the 30 career scenarios (draft labels, one sample). Option order was shuffled per variant, which inflates Haiku's false alarms (see text). †The Opus rerun stopped partway through when API credits ran out (18–19 scenarios per variant); a cell counts only scenarios where both the base and the variant were answered, so refusals (1–4 per variant) make some denominators 17.

irrelevant, subthreshold and neutral-mention controls and five option orders; we checked that our scenarios do not overlap Nimble's published training data. (3) A matched-order Claude rerun if API credit allows.

Resources and questions. We need two labelers who did not write the scenarios (we estimate two hours each) and Claude API credit (a matched-order rerun is 1,260 calls per sample); GPU time for step 2 is under $5. Is a 9B model plus the partial Claude results enough for the final report if no more credit is available?

6 Teamwork

Insiah Kizilbash wrote the career scenarios and their variants and built the threshold sweeps. Haiden Hunter built the analysis harness and agreement scoring and is organizing the blind labeling. Greg Betron ran the Claude baselines and the multi-turn pushback test. Shreyash Ranjan built the Nimble and Qwen pipeline on rented GPUs, ran the adapter comparison and wrote the report.

7 AI Usage

The team designed the study, wrote the scenarios, ran the experiments and decided every claim here. AI assistants helped as follows. **Code:** an AI coding assistant (Claude Code) wrote or revised much of the experiment, analysis and test code, which we reviewed and ran. **Auditing:** it recomputed our tables from raw results and audited the pipeline, finding bugs we fixed. **Writing:** it drafted and edited parts of this report, and ChatGPT reviewed drafts; we checked every number against the result files. **Scenarios:** written by the team; AI was used only to help edit wording. **Labels:** an AI-produced set of reference labels was not used, because it reproduced our draft labels; no AI output is used as a human label. The Claude, Qwen and Nimble models are subjects of the experiments, not assistants.

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